

MathCounts

This course aims to help the gifted 7th and 8th grade students who are interested in participating AMC 8 and Math counts competitions. Middle school students who are registering with their school math club are strongly encouraged to take this class. Other students with enthusiasm, dedication, concentration, and critical thinking will also be accepted. The critical yet interesting topics are selected, focusing on the subjects such as algebra, counting, probability, number theory and geometry. Problem solving strategy is emphasized throughout the class. Past AMC 8 and Mathcounts problems will be used as the homework.

Below are the selected topics from the AMC8/Mathcounts Class.

- Topic 1 Number Sense
- Topic 2 Finding Patterns
- Topic 3 Rate, Distance, and Time
- Topic 4 Counting: Permutation and Combination
- Topic 5 Probability
- Topic 6 Pythagorean Theorem
- Topic 7 Eight Ways to Calculate Triangle Area
- Topic 8 Area Method
- Topic 9 Pascal's Triangle
- Topic 10 Quadratic Equation
- Topic 11 Root Properties of Quadratic Equation
- Topic 12 Number Theory
- Topic 13 Common Algebra Tricks
- Topic 14 Mean, Median, Mode, and Range
- Topic 15 Pigeon Hole Principle
- Topic 16 Ratios and Proportions
- Topic 17 Exponents and Radicals
- Topic 18 Venn Diagram
- Topic 19 Arithmetic Sequence
- Topic 20 Geometric Sequence

■ 中学数学竞赛补习班/MathCounts

本课的目的在于利用带有趣味性及挑战性的数学问题来激发学生对数学的兴趣，并深化学生综合解题能力和提高解题速度。学生在课堂上遇到的数学问题将会是中学数学竞赛中常见的考题，难易兼而有之。教学方式为：学生完成课堂考试；教师分析试题并讲解解题技巧；学生完成家庭作业以巩固所学的知识。